A Novel Pixel Orientation Estimation Based Line Segment Detection Framework, and Its Applications to SAR Images

Chenguang Liu, Cuiling Liu, Chisheng Wang, Wu Zhu, and Qingquan Li

***Abstract*— In this article, we propose a new line segment detection framework based on a novel pixel orientation estimation (POE) method, which detects line segments from binary edge maps and can be combined with any edge detectors. We show its efficiency by testing it in SAR images. The proposed POE-based line segment detection framework aims to leverage the success of deep learning models for edge detection in 1-look SAR images. The novel POE method estimates the orientation of each edge pixel by counting the number of edge pixels along a set of orientations. As the most edge pixels exist along the orientation of the line segment, the orientation of the edge pixel is given by the orientation that gives the maximum number of counts. Counting the number of edge pixels along different orientations is equivalent to convolving the local neighbourhood of each edge pixel with a set of carefully designed window functions with each window function corresponding to a fixed orientation. With the estimated orientations of pixels in the edge map, pixels can be grouped with a region growing step to form line support regions. Regions with their size larger than a size threshold will be accepted. Finally, rectangles are used to approximate the accepted regions and those rectangles are detected line segments. Experiments in both simulated SAR dataset and real SAR images demonstrate the efficiency of the proposed method. In particular, we advance the state-of-the-art performances by 18% (F1-score) on the 1-look dataset simulated from YorkUrban-LineSegment Dataset.**

***Index Terms*— 1-look SAR images, binary edge maps, deep learning, line segment detection framework, pixel orientation estimation (POE).**

I. Introduction

LINE segments, which we define as straight parts of edges here, are important features of images. Line segment detection has been a long standing yet challenging problem in computer vision. Many methods have been proposed for line segment detection in optical images, including traditional detectors [1], [2], [3], [4], [5], [6] and deep learning-based methods [7], [8], [9], [10], [11], while there are much fewer line segment detectors dedicated to SAR images, mostly because of the strong multiplicative noise. Due to the strong differences in the statistics between optical and SAR images, methods developed for optical images are not suitable for detecting line segments in SAR images.

The classical ways to detect line segments in SAR images are based on Hough transform [1], [3], [12], [13], [14], [15], [16]. A ratio operator [17], [18] is first applied to detect the edges in SAR images, and then Hough transform is applied to detect lines in SAR images. Finally, line segments are obtained with a postprocessing step. More specifically, Hough transform tries to find the cues about the existence of a line segment by transforming the pixel coordinates into a parameter space. According to Hough transform, a line in the image space corresponds to a point in the parameter space. Therefore, the local maxima in the parameter space correspond to candidate lines. The maximum number of lines to be detected has to be set beforehand for Hough transform-based methods. In addition, local maxima with their number of pixels smaller than a given threshold will be discarded. Besides, two more thresholds are required to find the end points of line segments and set the length of the shortest line segment that is allowed. These thresholds usually have to be tuned according to the images at hand and it is difficult to find a set of threshold values that works globally. It can be seen from Fig. 1(e)–(f) that when the parameter settings change, line segment detection results computed by Hough transform may be quite different. Except for the threshold problem, Hough transform-based methods may suffer from the problem of multiple responses due to the digitization of pixel coordinates in images. This phenomenon is shown in Fig. 1(e)–(f) that multiple line segments are detected from the same location. The performances of Hough transform-based method can be improved by using the gradient magnitude field as input rather than using binary edge maps [19], [20], but the drawbacks of Hough transform remain.

A recent line segment detector for SAR images, which is named as LSDSAR [21], [22], has been shown to be able to detect line segments with a number of false detection control. An appealing point of this method is that it requires

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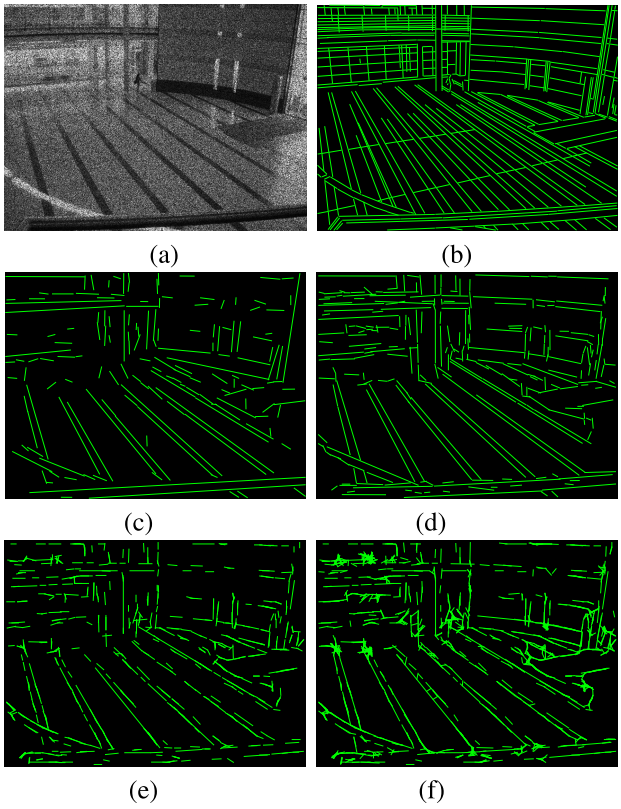


Fig. 1. Line segment detection results obtained by LSDSAR, Hough transform-based methods, and our POE-based method in a 1-look simulated SAR image. Size of the image is 480 × 640 pixels. Same edge detector is used for the Hough transform-based method and our POE-based method. For (e) and (f), the parameter settings for Hough transform are different. (a) 1-look image. (b) Ground truth. (c) LSDSAR (α = 4). (d) Proposed method. (e) Hough transform (setting 1). (f) Hough transform (setting 2).

few parameter tunings. LSDSAR is based on a Markovian a contrario model and the Helmholtz principle [23]. It first computes the gradient orientation of each pixel with a ratio-based gradient computation method [24], and then groups pixels sharing the same gradient orientation to get candidate line segments. Candidate line segments are finally validated by measuring the unlikeliness of a similar segment to appear by chance in a random image. The more unlikely a line segment is to appear by chance in a random image, the more likely it is to be a meaningful detection [25]. Even though LSDSAR has been proven to be effective to detect line segments in SAR images, its performances in 1-look SAR images are not satisfying enough. It can be seen from Fig. 1(c) that many true line segments are missing for LSDSAR.

On the other hand, deep learning techniques have been successfully used for edge detection in SAR images. A deep learning-based edge detector named GRHED [26] has been shown to be much more efficient for detecting edges in 1-look real SAR images while the performances of it are much less sensitive to the choice of parameter values (compared to the traditional ratio-based edge detectors for SAR images). GRHED is an edge detection framework constituted by a hand-crafted layer and fully convolutional layers, enabling models trained using datasets simulated from optical datasets to be efficient for edge detection in SAR images. Motivated by the success of applying deep learning-based methods for edge detection in 1-look real SAR images, we propose a new line segment detection framework, which detects line segments from binary edge maps, to improve line segment detection results in SAR images.

All line segment detectors need to group edge pixels belonging to the same line segment. Hough transform-based methods transform the coordinates of edge pixels into a parameter space to find the cues for the existence of line segments [1], [3]. Desolneux *et al.* [25] first compute the gradient orientation of each pixel and then search for meaningful line segments from all lines starting from the sides of images. Methods based on perceptual grouping [2], [4], [21] typically generated candidate line segments by grouping pixels sharing the same gradient orientation. In [6], edge pixels along the same line segment are grouped by exploiting the intrinsic property of line segments in image space, where line segments are represented by a set of linelets (a linelet is a set of either horizontally or vertically connected pixels). Despite the existence of these edge grouping methods, they cannot be used to detect line segments from the results computed by deep learning-based edge detectors except for Hough transform. Gradient orientations of pixels are used during either edge grouping or line segment validation process [2], [4], [6], [21], [25] while deep learning-based edge detector [26], [27], [28] computes only the edge strength map without the edge orientation information. Hough transform seems to be the best choice to detect line segments from binary edge maps (we do not discuss on those deep learning-based line segment detectors for two reasons: first, to the best of our knowledge, there is still no real SAR dataset for line segment detection in SAR images; second, most deep learning-based line segment detectors combined the detection of line segments and junctions generating candidate line segments by connecting two junctions [9], [10], [11]. Therefore, the strategies used in deep learning-based methods are not suitable for our situation).

Since Hough transform-based methods suffer from the problem of parameter tuning and multiple responses, we propose a brand-new edge grouping method based on a pixel orientation estimation (POE) method, which contributes to a line segment detection framework that can be combined with any edge detectors. More precisely, the orientation of edge pixels can be obtained by counting the number of edge pixels along a set of orientations. The orientation of the edge pixel is then given by the orientation that gives the maximum number of counts as most edge pixels exist along the line segment. Counting the number of edge pixels along different orientations is equivalent to convolving the local neighborhood centering at the pixel with a series of circle-shaped window functions (with each window function corresponding to a fixed orientation). By estimating the orientation of each edge pixel, we are able to obtain a set of line support regions with a region growing algorithm by grouping pixels having the same orientation. Generated regions will be checked by comparing their size with a threshold based on the Helmholtz principle [23].

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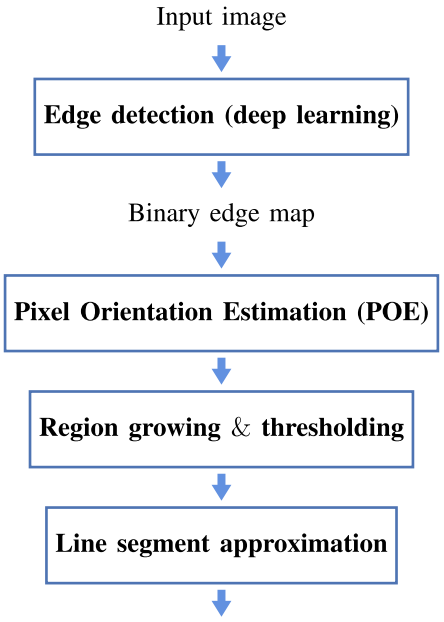


Fig. 2. Framework of the method.

Rectangles are finally used to describe accepted regions and those rectangles are detected line segments. The pipeline of the proposed line segment detection framework is shown in Fig. 2.

The contributions of this article can be summarized as follows.

1) We propose a brand-new edge grouping method based on a novel POE method, which enables us to generate line segments by grouping edge pixels with the estimated orientations. The resulting line segment detection framework requires few parameter tunings.
2) the proposed line segment detection framework can be easily combined with any edge detectors, which means that any improvements in the performances of edge detectors will contribute to a better line segment detector.
3) The proposed line segment detection framework advances the state-of-the-art performances by a large margin in 1-look SAR images.

The rest of this article is organized as follows: in Section II, we briefly introduce the edge detectors we use to detect edges of 1-look SAR images; in Section III, we will give details about our novel POE method, and introduce the region growing, region size thresholding, as well as the line segment approximation step; experiments in both 1-look simulated SAR dataset and three 1-look real SAR images will be provided in Section IV to demonstrate the efficiency of the proposed method; some conclusions will finally be given in Section V.

II. Edge Detection Using Deep Learning-Based Edge Detectors

In this article, we propose to follow the traditional fashion of detecting line segments in images by grouping edge pixels, based on a novel POE method. This is mainly to fully exploit the success of deep learning-based method for edge detection in SAR images [26]. Liu *et al.* [26] proposed a framework composed of a hand-crafted layer and fully convolutional layers, which aims at enabling models trained using simulated dataset to work in real SAR images. In the absence of a real SAR dataset for the task of edge detection, GRHED [26] proposed a solution to the application of deep learning models to real SAR images.

Contrarily to the traditional fully learnable deep models, there is a hand-crafted layer before learnable layers in GRHED. The hand-crafted layer, which is defined by a ratio-based gradient computation method GR [24], ensures that the distributions of feature maps computed by GR in simulated SAR and real SAR images are similar. Thus, the learnable layers trained on the feature maps of simulated SAR images will be able to perform efficient edge detection in the feature maps computed by GR in real SAR images.

In GRHED [26], the convolutional layers are identical to those in HED [27], [29] and this is the reason why it is called GRHED. However, GRHED is actually a framework, both the ratio operator (GR) and convolutional layers (HED) are replaceable (we can replace GR with any other ratio operators [17], [18] and replace HED layers with other convolutional layers). In this work, we also replace the convolutional layers with those of RCF [28] and keep the ratio operator GR intact. We call it as GRRCF. During training, the loss function of GRHED is identical to that in HED [27] and the loss function of GRRCF is identical to the one in RCF [28].

For the training of GRHED and GRRCF, we use two datasets. The first one is the same as the one used in [26], which is simulated by multiplying each image in the augmented BSDS500 [27], [30] with 1-look noise (it contains 28 800 images for training and validation). We train GRHED on this dataset and keep all the settings for training in [26], except that we set the training iterations to be 30 000 (more details about the training settings can be found in [26]). GRHED trained with this dataset is called GRHED. The other dataset is simulated from a dataset that mixes the augmented BSDS500 with the flipped VOC Context dataset [28], [31], [32], [33] (it contains 49 000 images in total), by multiplying each image with 1-look noise. Both GRHED and GRRCF will be trained using this dataset and we call them as GRHED-aug and GRRCF-aug, respectively. The feature maps computed by GR with multiple α values ($\alpha = 2, 3, 4, 5$, α is the exponential weight parameter used in the ratio operator GR) will be combined and used as the input of learnable layers, as done in [26].

Different from GRHED, which is implemented in Tensorflow, both GRHED-aug and GRRCF-aug are implemented in PyTorch. GRHED-aug and GRRCF-aug are trained from scratch with Xavier initialization [34]. Both models are trained using Adam optimizer with learning rate 0.0001. The batch size is set to 10 and the weight decay is set to 0.0002. GRHED is trained for 20 epochs and the learning rate will be divided by 10 after 30 000 iterations. GRRCF is trained for ten epochs (we tried to train GRRCF for 20 epochs and divide the learning rate by 10 after ten epochs, but it does not

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TABLE I
COMPARISON OF GRHED, GRHED-AUG, AND GRRCF-AUG ON 200 1-LOOK IMAGES SIMULATED FROM THE TESTING IMAGES OF BSDS500

Methods	ODS F1	OIS F1	AP
GRHED	0.667	0.685	0.690
GRHED-aug	0.694	0.710	0.724
GRRCF-aug	0.693	0.710	0.716

improve the performances). Nonmaxima suppression [35] and plain thresholding are applied to GRHED, GRHED-aug, and GRRCF-aug to obtain the binary edge map.

We compare the performances of GRHED, GRHED-aug, and GRRCF-aug on the 200 1-look images, which are simulated from BSDS500 dataset, in terms of three criteria: the optimal dataset scale (ODS) F1-score (one threshold for all the images), optimal image scale (OIS) F1-score (one threshold for each image), and average precision (AP). The results are shown in Table I.

From Table I for the comparison between GRHED and GRHED-aug, we can see that we push forward the ODS F1-score from 0.667 to 0.694 by using more training data. From the comparison between GRHED-aug and GRRCF-aug we can see that for these training settings, they show comparable detection capabilities, even though RCF [28] outperforms HED [29] in BSDS500 dataset (0.806 compared to 0.790). Conclusions may be different if we try more training settings for GRHED-aug and GRRCF-aug, but we leave it for future work as the main contribution of this work is that we propose a new line segment detection framework based on the POE method, which will enable us to detect line segments from binary edge maps produced by any edge detectors.

Edge maps computed by GRHED, GRHED-aug, and GRRCF-aug in a 1-look SAR image (simulated with an image from YorkUrban-LineSegment dataset) are shown in Fig. 3. Similar to the observation in Table I, GRHED-aug and GRRCF-aug detects slightly more true edge pixels than GRHED. In this work, we will detect line segments from binary edge maps (as those shown in Fig. 3) computed by GRHED, GRHED-aug, and GRRCF-aug, and compare the performances of the proposed methods with the state-of-the-art line segment detectors for SAR images.

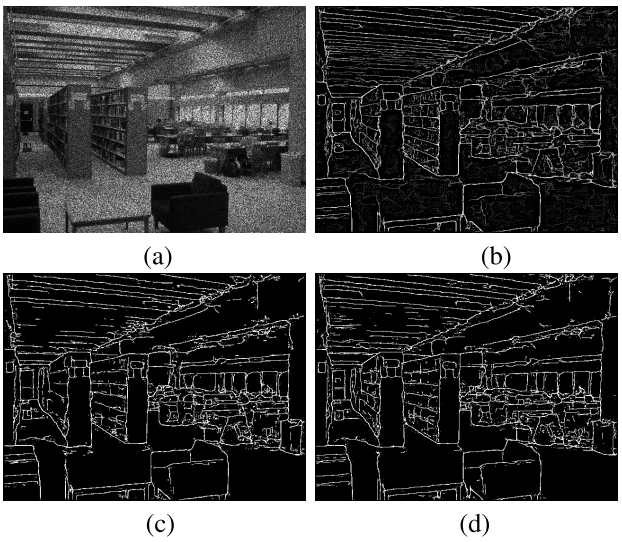


Fig. 3. Edge maps computed by GRHED, GRHED-aug, and GRRCF-aug in a 1-look simulated SAR image. Size of the image is 480 × 640 pixels. (a) 1-look image. (b) GRHED. (c) GRHED-aug. (d) GRRCF-aug.

III. LINE SEGMENT DETECTION WITH A NEW EDGE GROUPING METHOD BASED ON THE POE METHOD

Deep learning-based edge detectors compute only binary edge maps, while missing the information of edge orientations. Hough transform is the most commonly used method to detect line segments from binary edge maps. However, the drawbacks of Hough transform make it difficult to be used in general applications. A better edge grouping method is expected to detect line segments from binary edge maps.

Perceptual grouping methods, which typically use the edge orientation to group edge pixels, do not suffer from the problems faced by Hough transform. Candidate line segments are generated by searching and grouping pixels which have the same edge orientation, and the end points of candidate line segments are naturally identified. The aim of this article is to propose a POE method which estimates the edge orientation of pixels from binary edge maps. The proposed method should force edge pixels along the same line segment to have the same orientation, and output different angles for pixels belonging to different line segments. Candidate line segments can then be generated through a region growing algorithm. In the following, we will first describe the novel POE method we propose in this work and give details about the subsequent procedures of the line segment detection framework, which includes region growing, region size thresholding, and line segment approximation.

A. Pixel Orientation Estimation

In this work, we propose a novel POE method using contextual information of each pixel in the binary edge map. Consider the binary edge maps shown in the second column of Fig. 4, it is natural that the orientation of an edge pixel should be aligned with the orientation of the line segment passing through it. Besides, all edge pixels along a line segment should have the same orientation. Estimating the orientation of an edge pixel can thus be transformed as estimating the orientation of the line segment passing through it. In order to find the orientation of the line segment passing through an edge pixel, we can count the number of edge pixels along a series of orientations. The maximum number of counts should appear when the orientation coincides with the orientation of the line segment. Therefore, the orientation of the line segment passing through an edge pixel can be found by counting the number of edge pixels toward a set of orientations. The orientation of the line segment is given by the orientation that gives the maximum number of counts.

More formally, counting the number of edge pixels along a set of orientations is equivalent to computing the convolution between the local region of an edge pixel with a set of carefully

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designed window functions. For an edge pixel $u(x, y)$ which locates at (x, y) in the binary edge map (u) of an SAR image, we consider a circle-shaped region which centers at (x, y), and compute the convolution of this region with a set of circle-shaped window functions respectively (with each window function corresponding to a fixed orientation), the orientation corresponding to the window function which gives the maximum response will be considered as the orientation of the edge pixel.

Assuming that we have P window functions, and the pixel values of these windows are all zero except for one diameter of each window. Pixel values along the exclusive diameter of each window are 1. The angles of these diameters are uniformly distributed over $[0, \pi)$ and the orientation of each window is defined as the orientation of its exclusive diameter. For the ith window $[i = 1, 2, \ldots, P$, defining the angle of its exclusive diameter θ_i as $((i-1) * \pi/P)]$, then the convolution between the circle-shaped region centering at pixel (x, y) and the ith window can be computed as

$$S_i(x, y) = \sum_{x'=-w}^{w} \sum_{y'=-w}^{w} u\left(x + x', y + y'\right) * f\left(x', y'\right) \quad (1)$$

where w is the radius of the circle-shaped region and circle-shaped windows, and

$$f\left(x', y'\right) = \begin{cases} 1.0, & \text{if } |-x'\sin(\theta_i) + y'\cos(\theta_i)| \\ & < 0.5 \text{ and } x'^2 + y'^2 <= w^2 \\ 0.0, & \text{otherwise.} \end{cases} \quad (2)$$

Assuming that

$$S_j(x, y) = \max_{1 \le i \le P} S_i(x, y) \quad (3)$$

then the edge orientation of pixel (x, y) is defined as

$$O(x, y) = \frac{(j-1) * \pi}{P} \quad (4)$$

which corresponds to the direction of the diameter of the jth window.

One important point here is that we need to choose an appropriate number of windows (the value of P). Several factors have to be taken into account: first, the number of windows should be large, so that we can distinguish two line segments with close orientations; second, because of the digitization of pixel location in the image space, when the number of directions is too large, two windows with their diameters orienting to two adjacent directions may overlap significantly; third, with the increase of the number of directions, the computation load increases rapidly. In our experiments, we found 16 directions to be a nice choice. It means that for each window, the angle of its diameter is one of the following values: 0, $(\pi/16)$, $(\pi/8)$, $(3\pi/16)$, ..., and $(15\pi/16)$. The other variable that matters here is the radius value w of these circle-shaped windows. We found the value of 10 appropriate as with a larger radius, the estimation of orientations is more stable for each edge pixel. The settings of these parameter values will be discussed further in the experimental section.

The 16 windows with radius size $w = 10$ are shown in Fig. 5. It is clear that convolving the binary edge maps as shown in the second column of Fig. 4 with the window functions shown in Fig. 5 is equivalent to counting the number of edge pixels along a set of orientations. Estimating the orientation of an edge pixel (or equivalently estimating the orientation of the line segment passing through it) can thus be transformed as finding the direction that gives the maximum response. This new method, which is named as POE is designed to estimate the angles of edge pixels from binary edge maps. Based on the estimated angles, line segments can be identified by grouping edge pixels sharing the same orientation. To show the efficiency of POE, we display the histograms of the estimated angles using POE in the third column of Fig. 4, where the angles are estimated from the binary edge maps shown in the second column of Fig. 4. It can be easily observed from the first three rows of Fig. 4 that nearly all edge pixels have the same angle as the line segments in those images are parallel. Although the angles of some edge pixels differ from the most frequent one, the angle difference is pretty small [just $(\pi/16)$]. Besides, it can be seen from the last row of Fig. 4 that as line segments with two different angles exist, two peaks appear in the histogram of the estimated angles. Line segments detected by grouping the edge pixels having the same estimated angle up to an angle tolerance $(\pi/16)$ are shown in the last column of Fig. 4. It is clearly shown that nearly all line segments in the binary edge maps are perfectly identified. In the following, we will describe the region growing algorithm, region size thresholding, and subsequent line segment approximation step.

B. Region Growing and Region Size Thresholding

After estimating the orientation of each edge pixel in the edge map, we use region growing method to group edge pixels. In other words, we will group adjacent pixels whose edge orientations are the same up to a certain angle tolerance τ $[\tau = (\pi/P)$ in our experiments]. The region growing step will go through all the edge pixels in the edge map. More precisely, we start from one edge pixel and use it as the seed pixel of a region, then we search in its 3×3 neighborhood, if the angle difference between one pixel and the seed pixel is not larger than τ, we will add this pixel into the region. Each time a pixel is added into the region, the searching process will loop over the 3×3 neighborhoods of all the pixels in the region searching for the pixels having the same angle as the seed pixel, until no more pixels can be added into the region. At the end of one searching process, we will check the size of the region (the number of pixels in the region). If the number of pixels in the region is larger than λ, it will be accepted and all the pixels will be labeled as USED so that they will not be visited again. The region will be rejected otherwise and the pixels in the region are released so that they can be added to other regions. The region growing step then starts from another edge pixel until it goes through all the edge pixels in the edge map.

The natural question is how to set the region size threshold λ. We set this threshold according to the Helmholtz

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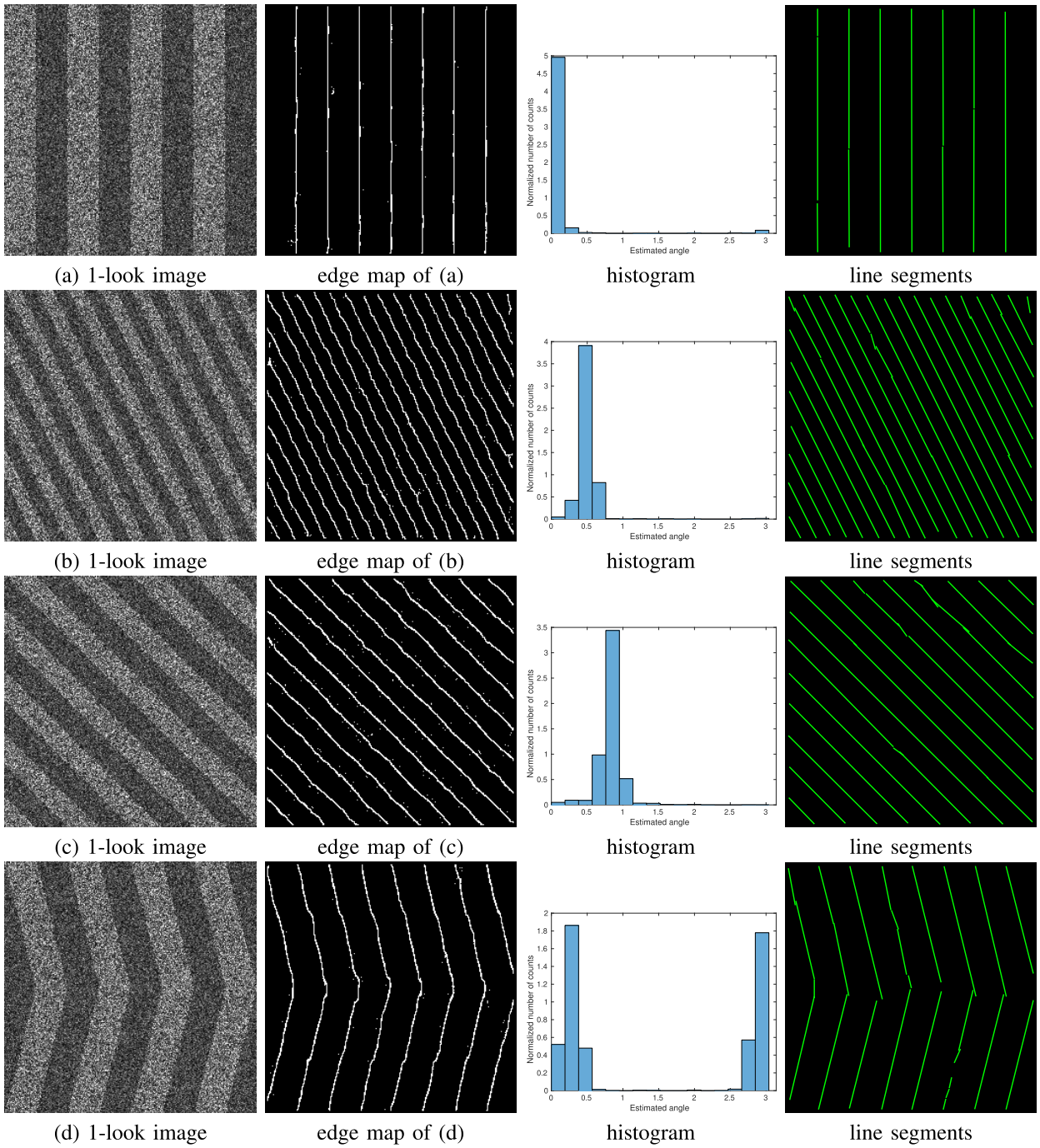


Fig. 4. Histograms of angles (third column) estimated by the proposed POE method from binary edge maps (second column) computed by GRHED-aug in several 1-look synthetic edge images (first column). In all the synthetic edge images, the amplitude ratio contrast of the edge is 1.6. Size of the images is 256×256 pixels. Line segment detection results computed by the proposed POE-based method are shown in the last column.

principle, which claims that no meaningful structure should be observed by chance in a random image. More precisely, according to the definition in [25], a line segment is meaningful if the expected number of its occurrences in a random image is small. The meaningfulness of a line segment is checked through the use of an a contrario model. According to the usual a contrario models [4], [21], [25], a line segment is considered to be ε-meaningful if its NFA is smaller than ε. Computing the NFAs to check the meaningfulness of a line segment implies a minimum length of line segment [25]. Here, we use the hypothesis in the usual a contrario model [4], [25]: first, the orientation of each pixel follows a uniform distribution over $[0, \pi)$ (with minor modifications as it is $[0, 2\pi]$ in the original version): second, the orientations of pixels are independent, random variables. Then, for the minimum meaningful line segment r with size $l_{\min}$, all pixels should have the same orientation as the line segment, in an image u of size (M, N), its number of false alarms should

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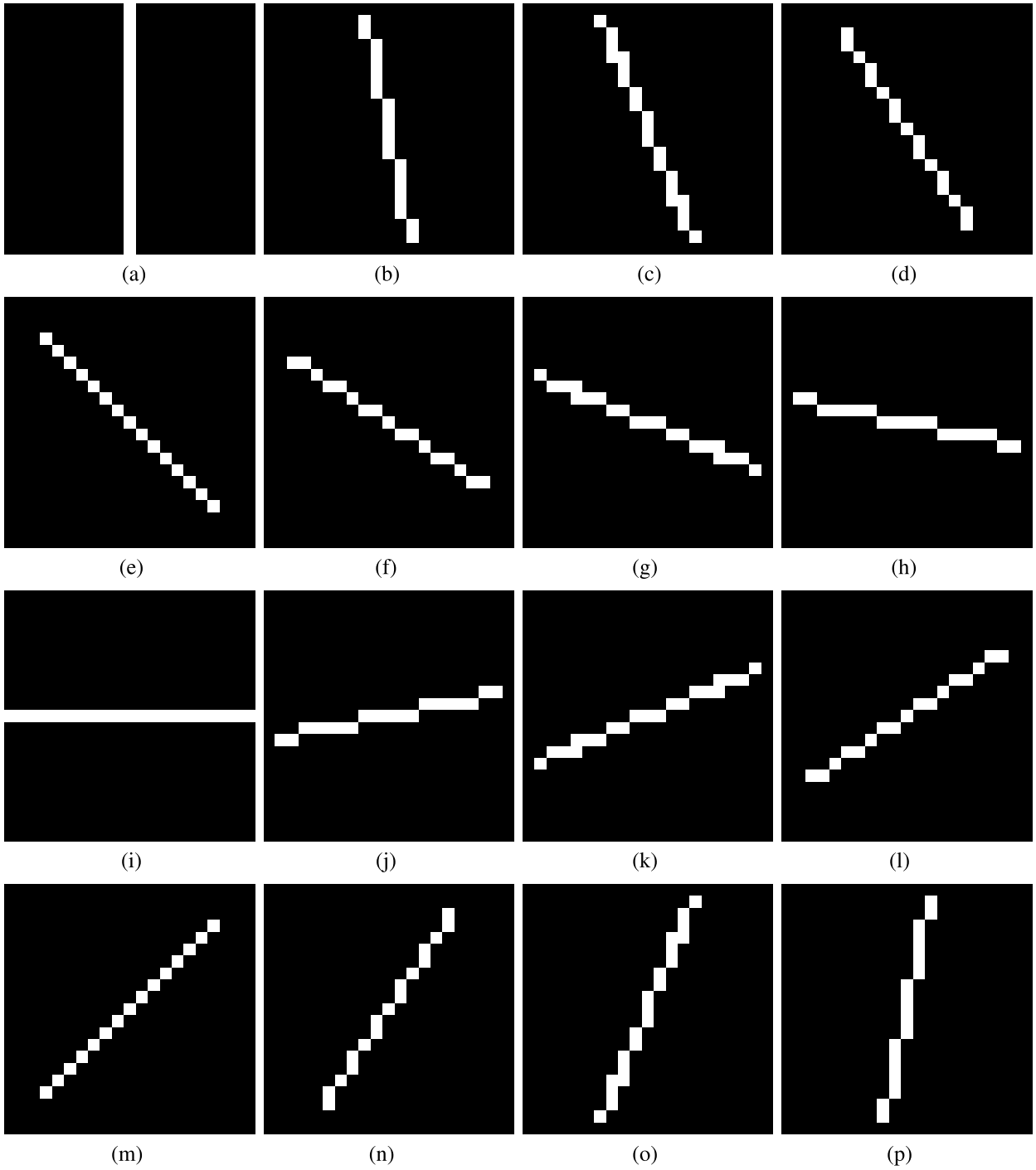


Fig. 5. Sixteen window functions with their diameters orienting to a certain direction. Radius of the windows is 10 pixels. The angle of the diameter for a window is one of the following: 0, $(\pi/16)$, $(\pi/8)$, $(3\pi/16)$, ..., and $(15\pi/16)$. (a) Angle: 0. (b) Angle: $(\pi/16)$. (c) Angle: $(\pi/8)$. (d) Angle: $(3\pi/16)$. (e) Angle: $(\pi/4)$. (f) Angle: $(5\pi/16)$. (g) Angle: $(3\pi/8)$. (h) Angle: $(7\pi/16)$. (i) Angle: $(\pi/2)$. (j) Angle: $(9\pi/16)$. (k) Angle: $(5\pi/8)$. (l) Angle: $(11\pi/16)$. (m) Angle: $(3\pi/4)$. (n) Angle: $(13\pi/16)$. (o) Angle: $(7\pi/8)$. (p) Angle: $(15\pi/16)$.

satisfy [4], [25]

$$\mathrm{NFA}(r) = N_R * p^{l_{\min}} \leq \varepsilon \tag{5}$$

where N_R represents the total number of possible line segments in a random image, and p represents the probability that a pixel has the same angle as the line segment. In our case $N_R = ((MN)^5)^{1/2}$ as done in [4] and [21]. As the angle tolerance τ is set to $(\pi/16)$, $p = (3/P)$ because of the uniform distribution hypothesis.

Then we have

$$l_{\min} \geq \frac{\log(\varepsilon) - \log(N_R)}{\log(p)}. \tag{6}$$

Therefore, only regions with their size larger than $(\log(\varepsilon) - \log(N_R)/\log(p))$ could be corresponding to a meaningful line segment. We set $\lambda = (\log(\varepsilon) - \log(N_R)/\log(p))$ and keep only those regions with their size (number of pixels) larger than $(\log(\varepsilon) - \log(N_R)/\log(p))$ (letting $\varepsilon = 1$ as done in [4], [21], and [25]).

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C. Line Segment Approximation

After the region growing and region size thresholding step, we have a set of regions which correspond to detected line segments. Here, we use rectangles to approximate those regions and the rectangles are considered to be detected line segments, as done in [4] and [21]. Four parameters are used to describe the rectangle: center, angle, length, and width.

We first describe the way to compute the center location of the region. Given a region, let $(x(i), y(i))$ be the coordinates of the ith pixel in the region, then the coordinates of the center pixel (c_x, c_y) can be computed as

$$c_x = \frac{\sum_{i \in \text{region}} x(i)}{N_{\text{region}}} \tag{7}$$

$$c_y = \frac{\sum_{i \in \text{region}} y(i)}{N_{\text{region}}} \tag{8}$$

where N_{region} is the number of pixels in the region.

Even though we have estimated the edge orientation of each pixel, and assume that their orientations are the same as the line segment, their orientations are not accurate enough. We compute the angle of the line segment as the angle between the horizontal axis and the first inertia axis of the region. The angle of the line segment can be computed as the angle of the eigenvector corresponding to the smallest eigenvalue of the following matrix:

$$\mathbb{M} = \begin{pmatrix} \mathbb{M}^{xx} & \mathbb{M}^{xy} \\ \mathbb{M}^{xy} & \mathbb{M}^{yy} \end{pmatrix} \tag{9}$$

where

$$\mathbb{M}^{xx} = \frac{\sum_{i \in \text{region}} (x(i) - c_x)^2}{N_{\text{region}}} \tag{10}$$

$$\mathbb{M}^{yy} = \frac{\sum_{i \in \text{region}} \big(y(i) - c_y\big)^2}{N_{\text{region}}} \tag{11}$$

$$\mathbb{M}^{xy} = \frac{\sum_{i \in \text{region}} (x(i) - c_x) \cdot \big(y(i) - c_y\big)}{N_{\text{region}}}. \tag{12}$$

The length and width of the rectangle are then selected as the smallest values that can cover the considered region.

IV. Experiments

In this section, we will demonstrate the efficiency of our proposed line segment detection framework by comparing it with the Hough transform-based methods and the state-of-the-art line segment detector LSDSAR [21] in both 1-look simulated SAR dataset and several 1-look real SAR images. We focus on the 1-look situation as it is the most challenging case for line segment detection in SAR images. As we use GRHED, GRHED-aug, and GRRCF-aug to compute the edge maps of images, we name the combination of these edge detectors with our POE-based line segment detection framework as GRHED-POE, GRHED-aug-POE, and GRRCF-aug-POE. We will also term them as POE-based line segment detectors in the following. The efficiency of the proposed line segment detection framework will be demonstrated by these three line segment detectors.



(a) (b)

Fig. 6. Image from YorkUrban-LineSegment dataset and the 1-look version (multiply the image with 1-look noise) of it. (a) Clean image. (b) 1-look version.

A. Comparison in 1-Look Simulated SAR Dataset

In order to compare different line segment detectors quantitatively in complex situations, we created a 1-look simulated SAR dataset using YorkUrban-LineSegment dataset proposed in [6], by multiplying each image in the dataset with 1-look noise. YorkUrban-LineSegment dataset is created by three annotators [6] who spent around two months labeling 102 images of size 480 × 640 pixels (45 indoor images, 57 outdoor images). These images are taken from scenes of York University campus and downtown Toronto, Canada [36]. As an example to show how difficult it is to detect line segments from 1-look images, we display an image from YorkUrban-LineSegment dataset and the 1-look version of it in Fig. 6. Our aim is to detect line segments in images with noise level similar to the one in Fig. 6(b).

1) Evaluation Criterion: We compare the performances of different methods in terms of F1-scores by transforming line segments into a binary line segment map, where pixel values are one if they belong to a certain line segment, and zero otherwise. The F1-score is computed as follows:

$$F1 = \frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}} \tag{13}$$

where precision is the fraction of correctly detected line segment pixels among all detected line segment pixels, while recall is the fraction of correctly detected line segment pixels among all the line segment pixels in the ground truth.

2) Comparison Between Hough Transform-Based Methods and Our POE-Based Methods: In this section, we compare our POE-based methods with Hough transform-based methods to demonstrate the efficiency of our new edge grouping method. Line segments will be detected by Hough transform and our POE-based method from edge maps computed by GRHED-aug. The combination of GRHED-aug with Hough transform will be called as GRHED-aug-Hough. Before comparing the performances of Hough transform-based methods with our POE-based methods, we first evaluate the influences of parameter settings on the performances of Hough transform-based methods and POE-based methods.

a) The influence of parameter settings on the performances of Hough transform-based methods

For Hough transform, four parameters have to be set by users as follows:

1) t_1, the number of peaks (or lines) to be detected;

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TABLE II
FIVE SETS OF PARAMETER VALUES USED FOR HOUGH TRANSFORM

Methods	t_1	t_2	t_3	t_4
GRHED-aug-Hough-1	5000	18	1	13
GRHED-aug-Hough-2	5000	18	3	13
GRHED-aug-Hough-3	5000	18	5	13
GRHED-aug-Hough-4	5000	18	5	18
GRHED-aug-Hough-5	5000	30	5	13

TABLE III
F1-SCORES COMPUTED BY HOUGH TRANSFORM IN THE SIMULATED SAR DATASET WITH DIFFERENT PARAMETER SETTINGS

Methods	F1-score ($\rho = 0.2$)	F1-score ($\rho = 0.4$)
GRHED-aug-Hough-1	0.2064	0.0.2031
GRHED-aug-Hough-2	0.5430	0.5283
GRHED-aug-Hough-3	0.6646	0.6445
GRHED-aug-Hough-4	0.6220	0.5922
GRHED-aug-Hough-5	0.6656	0.6442

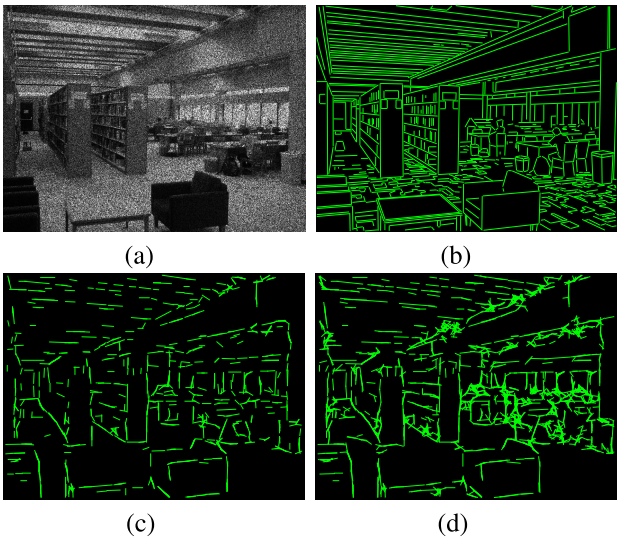


Fig. 7. Line segment detection results obtained by Hough transform-based method with two sets of parameter values in a 1-look simulated SAR image. Size of the image is 480 × 640 pixels. (a) 1-look image. (b) Ground truth. (c) GRHED-aug-Hough-2. (d) GRHED-aug-Hough-5.

2) t_2, the smallest number of pixels in a bin, or in other words, the smallest number of pixels in a line;
3) t_3, the minimum gap between two line segments along a line; and
4) t_4, the length of the shortest line segment that is allowed.

As no default setting is proposed for the parameters in Hough transform, we choose five sets of values as displayed in Table II to analyze the influence of parameter settings on the performances of Hough transform-based method. t_1 is always set to 5000 as it corresponds to the maximum number of lines to be detected. By setting it to be 5000, we aim to detect all possible lines in the image.

F1-scores computed by Hough transform for each set of parameter values in the simulated SAR dataset are displayed in Table III. From the comparison between the second column and third column of Table III we can see that the setting of ρ does not influence much on the performances of Hough transform (ρ is used to binarize the edge strength map computed by GRHED-aug). F1-scores obtained by Hough transform with $\rho = 0.2$ is generally 2% or 3% higher than those obtained by Hough transform with $\rho = 0.4$. Besides, it can be easily observed from the first three rows of Table III that the performances of Hough transform is very sensitive to the setting of t_3, which corresponds to the minimum gap between two line segments in the same bin. It is clearly shown that with a minor change of t_3, F1-scores obtained by Hough transform change a lot. From the last three rows of Table III, we can see that the settings of t_2 and t_4 are of minor influence on the performances of Hough transform.

From the observation in Table III, we can see that setting 3 and setting 5 seem to be good choices for Hough transform. However, when we visualize the line segment detection results shown in Fig. 7, we find that many false line segments are detected by GRHED-aug-Hough-5. Besides, the phenomenon that several line segments are detected from the same location appears frequently. Similar problems have been observed for GRHED-aug-Hough-3. This is probably because the value of t_3 is set too large (with $t_3 = 5$), which means that for pixels along a line, they will be connected to form a line segment if their distance is not larger than 5 pixels, while they are assumed to belong to different line segments if their distance is larger than 5 pixels. Line segment detection results computed by GRHED-aug-Hough-2 (with $t_3 = 3$) seems to be better as much fewer false detections are produced, even though the F1-score computed by it is approximately 12% lower than that computed by GRHED-aug-Hough-3 and GRHED-aug-Hough-5. The problem of multiple responses is also relieved when t_3 is set to 3.

Observing the performances of Hough transform-based method with different parameter settings as shown in Table III and Fig. 7, we will use the settings as displayed in the second row and third row of Table II. Therefore, we will compare the performances of GRHED-aug-Hough-2 and GRHED-aug-Hough-3 with other methods.

b) The influences of parameter settings on the performances of our POE-based methods

For our POE-based line segment detectors, three parameters have to be set:

1) w: the radius of the circle-shaped windows;
2) P: the number of orientations; and
3) τ: the angle tolerance used in the region growing step.

We first evaluate the influence of the radius size on the performances of our POE-based method in the simulated SAR dataset. During the evaluation, the other parameters are fixed with $\rho = 0.4$, $P = 16$, and $\tau = (\pi/16)$. F1-scores computed by GRHED-aug-POE with different settings of radius size are shown in Table IV. It is clearly shown in Table IV that the influences of radius size on the performances of our POE-based method are minor.

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TABLE IV

F1-SCORES COMPUTED BY GRHED-AUG-POE WITH DIFFERENT RADIUS SIZE. ρ IS SET TO 0.4, P IS SET TO 16 AND τ IS SET TO $(\pi/16)$

radius size (w)	5	8	10	12	15
F1-score	0.6068	0.6122	0.6178	0.6206	0.6216

TABLE V

F1-SCORES COMPUTED BY GRHED-AUG-POE ON THE SIMULATED SAR DATASET WITH DIFFERENT SETTINGS OF P. ρ IS SET TO 0.4, τ IS SET TO $(\pi/16)$ AND THE RADIUS SIZE w IS SET TO 10 PIXELS

Number of Orientations	8	16	32
F1-score	0.5717	0.6178	0.6149

The influences of the setting of P on the performances of the POE method are further evaluated with $\rho = 0.4$, $\tau = (\pi/16)$ and the radius size $w = 10$. F1-scores computed by GRHED-aug-POE with different settings of P is displayed in Table V. From the F1-scores displayed in Table V we can see that with different settings of P, F1-scores obtained by GRHED-aug-POE do not change a lot. From the comparison between the third column and fourth column of Table V, we can see that the performances of GRHED-aug-POE are not improved by considering a larger number of orientations. From the comparison between the second column and third column of Table V, we can see that the performances of GRHED-aug-POE decrease with smaller number of orientations considered. Two line segments with their angle close to each other may not be distinguished by considering a smaller number of orientations. In addition, when P is set to 8 and τ is set to $(\pi/16)$, the POE method groups only pixels with exactly the same angle, while for $P = 16$ and $\tau = (\pi/16)$, pixels with their angle differences no larger than $(\pi/16)$ will be gathered together. This could be another reason for the decrease of performances of GRHED-aug-POE with $P = 8$.

The influences of the setting of τ on the performances of GRHED-aug-POE are evaluated in the simulated SAR dataset as shown in Table VI. It is clearly shown in Table VI that F1-scores obtained by GRHED-aug-POE become higher when a larger value of τ is used. The F1-score obtained by GRHED-aug-POE with $\tau = (\pi/32)$ is more than 10% lower than GRHED-aug-POE with other settings of τ. This could be because, with this setting of τ, only pixels with exactly the same angle will be grouped. Even though GRHED-aug-POE with $\tau = (\pi/8)$ obtains the highest F1-score, the setting of $\tau = (\pi/16)$ is preferred as when τ is set to $(\pi/8)$, GRHED-aug-POE may not be able to distinguish two line segments with their angle difference smaller than $(\pi/8)$.

From the observations in Tables IV–VI, we propose to use the following setting as the default setting of our POE-based method: the radius size w of the circle-shaped windows is set to 10 pixels, the number of orientations P is set to 16, and the angle tolerance τ is set to $(\pi/16)$.

For a visual inspection, we compare the performances of GRHED-aug-Hough-2 and GRHED-aug-POE with the prosed setting in a 1-look simulated SAR image as shown in Fig. 8.

TABLE VI

INFLUENCES OF THE SETTING OF τ ON THE PERFORMANCES OF GRHED-AUG-POE. RADIUS OF THE CIRCLE-SHAPED WINDOWS IS SET TO 10 PIXELS, ρ IS SET TO 0.4 AND THE NUMBER OF ORIENTATIONS P IS SET TO 16

τ	$\frac{\pi}{8}$	$\frac{\pi}{16}$	$\frac{\pi}{32}$
F1-score	0.6406	0.6178	0.4957

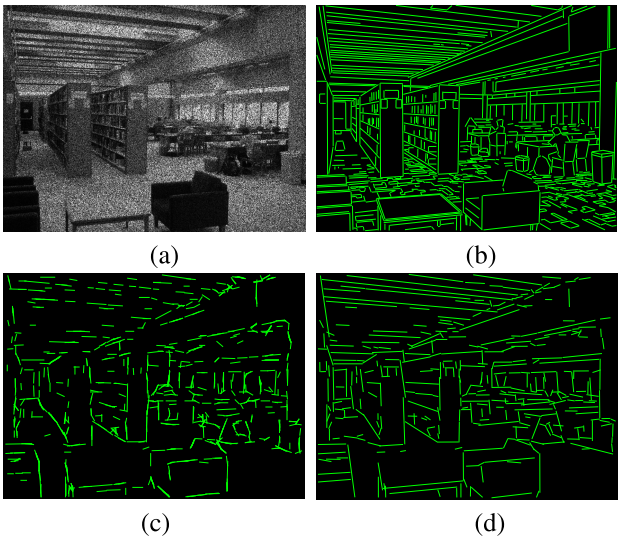


Fig. 8. Line segment detection results obtained by GRHED-aug-Hough-2 and GRHED-aug-POE in a 1-look simulated SAR image. Size of the image is 480 × 640 pixels. (a) 1-look image. (b) Ground truth. (c) GRHED-aug-Hough-2. (d) GRHED-aug-POE.

It is clearly shown in Fig. 8 that GRHED-aug-POE detects much more true line segments than GRHED-aug-Hough-2, while fewer false detections are produced by our POE-based method. Besides, line segments detected by GRHED-aug-Hough-2 tend to be fragmented. The completeness of line segments detected by Hough transform-based methods can be improved with a larger value of t_3, as displayed in Fig. 7(d), but more false detections are produced. Furthermore, it can be seen from both Figs. 7 and 8 that the problem of multiple responses exists for Hough transform-based methods.

From all the experiments above, the advantages of our POE-based methods over Hough transform-based methods are clear.

1) Parameter values for our POE-based methods are much easier to choose, and the performances of our POE-based methods are much less sensitive to the choice of parameter values.
2) The problem of multiple responses exist for Hough transform-based methods, but it is not the case for our POE-based methods.
3) The proposed POE-based methods detect more true line segments while producing fewer false detections, the line segments detected by our POE-based method are more complete.

More comparisons will be done between Hough transform-based methods and our POE-based methods

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in the following to show the efficiency of our new edge grouping method based on POE.

3) Comparison Between LSDSAR and Our POE-Based Methods: In this part, we compare the performances of our POE-based methods with the state-of-the-art line segment detector for SAR images LSDSAR.

a) Parameter settings: For our POE-based line segment detectors, the radius of the circle-shaped windows is set to 10 pixels, the number of orientations is set to 16, the angle tolerance τ is set to $(\pi/16)$, the region size threshold λ is automatically computed according to the Helmholtz principle with $\varepsilon = 1$.

For deep learning-based edge detectors that we will use to compute the binary edge maps, namely GRHED-POE, GRHED-aug-POE, and GRRCF-aug-POE, one important factor that may influence the performances of the resulting detectors is the threshold ρ used to binarize the edge strength map during plain thresholding. We vary this threshold ρ from 0.1 to 0.5 with step 0.1 for all the deep learning-based edge detectors and test its influences on the F1-scores for each method (the range of pixel values in the edge probability maps computed by deep learning-based methods is [0, 1]).

For LSDSAR, we vary the exponential weight parameter α used in the gradient computation step (GR) from 1 to 5 with step 1 and keep the other parameter settings unchanged [21]. We compute F1-scores for LSDSAR with each α value.

b) Evaluation results: The F1-scores computed by LSDSAR, GRHED-POE, GRHED-aug-POE, and GRRCF-aug-POE in the 1-look simulated SAR dataset are shown in Table VII. The best F1-score obtained by each method is highlighted in larger bold font.

TABLE VII

F1-SCORES COMPUTED FOR LSDSAR WITH DIFFERENT α VALUES AND THE POE BASED LINE SEGMENT DETECTORS WITH DIFFERENT SETTINGS OF ρ. ρ IS USED TO BINARIZE THE EDGE STRENGTH MAP (ESM) COMPUTED BY DEEP LEARNING-BASED EDGE DETECTORS. RANGE OF PIXEL VALUES IN THE ESM IS [0.0, 1.0])

Methods	F1-score	Methods	F1-score
LSDSAR ($\alpha = 1$)	0.2160	GRRCF-aug-POE ($\rho = 0.1$)	**0.6556**
LSDSAR ($\alpha = 2$)	0.4001	GRRCF-aug-POE ($\rho = 0.2$)	0.6429
LSDSAR ($\alpha = 3$)	0.4616	GRRCF-aug-POE ($\rho = 0.3$)	0.6161
LSDSAR ($\alpha = 4$)	**0.4810**	GRRCF-aug-POE ($\rho = 0.4$)	0.5695
LSDSAR ($\alpha = 5$)	0.4742	GRRCF-aug-POE ($\rho = 0.5$)	0.4962
GRHED-POE ($\rho = 0.1$)	0.6497	GRHED-aug-POE ($\rho = 0.1$)	**0.6677**
GRHED-POE ($\rho = 0.2$)	**0.6546**	GRHED-aug-POE ($\rho = 0.2$)	0.6623
GRHED-POE ($\rho = 0.3$)	0.6451	GRHED-aug-POE ($\rho = 0.3$)	0.6459
GRHED-POE ($\rho = 0.4$)	0.6216	GRHED-aug-POE ($\rho = 0.4$)	0.6178
GRHED-POE ($\rho = 0.5$)	0.5870	GRHED-aug-POE ($\rho = 0.5$)	0.5690

From Table VII, we can see that the highest F1-score obtained by LSDSAR is 0.4810 with $\alpha = 4$. It is much lower (more than 10%) than the other methods, which shows the efficiency of the proposed line segment detection framework. In addition, GRHED-aug-POE with $\rho = 0.1$ advances the performances of the state-of-the-art LSDSAR with $\alpha = 4$ by approximately 18% (0.6677 compared to 0.4810). An interesting phenomenon is that with the value of ρ increasing, it seems that the detection capabilities of GRHED-POE, GRHED-aug-POE, and GRRCF-aug-POE become weaker. The main reason is that we may discard true edge pixels when we use a larger value of ρ, even though we detect fewer noise pixels with it. It can also be seen from Table VII that with all settings of ρ, the F1-scores obtained by POE-based line segment detectors are almost always much higher than the best F1-score obtained by LSDSAR.

Although the POE-based line segment detectors with $\rho = 0.1$ or 0.2 seem to have better detection capability, more false line segments could be detected as more false edge pixels will be produced by deep learning-based edge detectors. It has been shown in [26] that the performances of deep learning-based edge detectors are not sensitive to the choice of ρ. In general applications, we propose to set ρ to 0.4 so that the edge maps computed by deep learning-based edge detectors are cleaner.

4) Line Segment Detection Results Computed by Hough Transform-Based Methods, LSDSAR and the POE-Based Methods in Two 1-Look Simulated SAR Images: For a visual comparison, we show the line segment detection results obtained by GRHED-aug-Hough-2 ($\rho = 0.4$), GRHED-aug-Hough-3 ($\rho = 0.4$), LSDSAR ($\alpha = 4$), GRHED-POE ($\rho = 0.4$), GRRCF-aug-POE ($\rho = 0.4$), and GRHED-aug-POE ($\rho = 0.4$) in two 1-look simulated SAR images (one indoor image and one outdoor image) in Figs. 9 and 10. It is clearly shown from the second row and third row of both figures that the proposed POE-based line segment detectors detect much more true line segments than LSDSAR while they produce only a few false detections. In both figures, the F1-scores computed by GRHED-POE and GRHED-aug-POE are at least 12% higher than that of LSDSAR. GRRCF-aug-POE achieves lower F1-scores than GRHED-POE and GRHED-aug-POE, but its F1-scores are still much higher than those of LSDSAR. Compared to the line segments detected by the POE-based methods, the line segments detected by LSDSAR tend to be fragmented. For the comparison among those POE-based methods, the performances of GRHED-POE and GRHED-aug-POE are similar, while both of them detect more true line segments than GRRCF-aug-POE. This conforms with the observation in Table VII that GRRCF-aug-POE obtains the lowest F1-score with $\rho = 0.4$.

From the comparison among Fig. 9(f)–(h), we can see that with the edge maps computed by the same edge detector, the proposed POE-based method obtains much better

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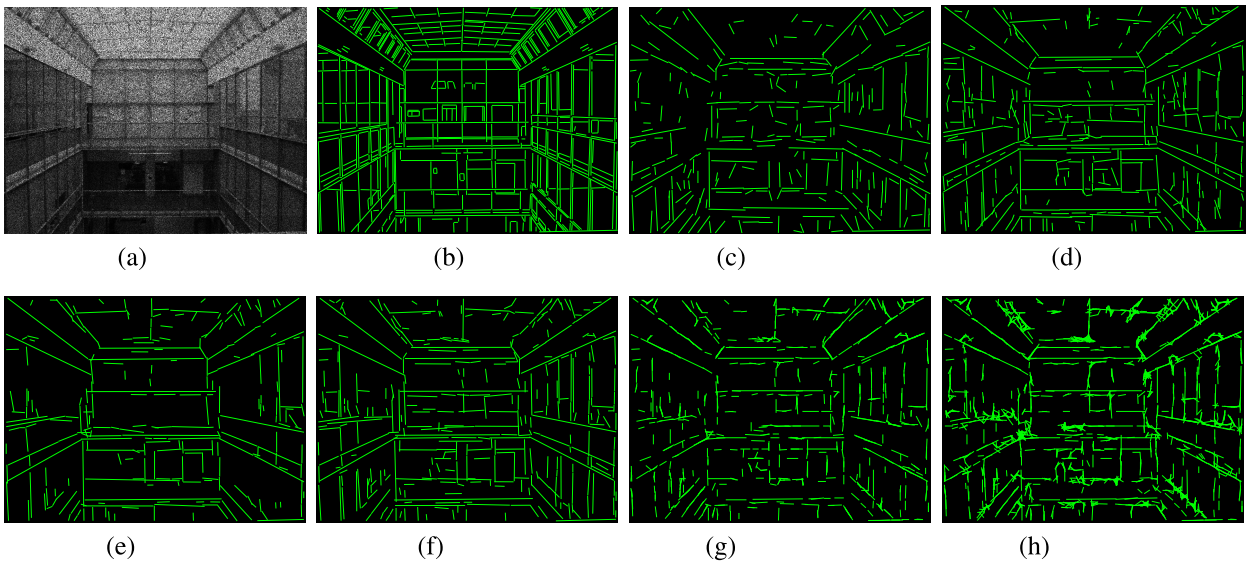


Fig. 9. Line segment detection results obtained by Hough transform-based method, LSDSAR and POE-based line segment detectors in a 1-look simulated SAR image (indoor). The size of the image is 480 × 640 pixels. (a) 1-look image. (b) Ground truth. (c) LSDSAR ($\alpha = 4$), F1-score: 0.566. (d) GRHED-POE, F1-score: 0.688. (e) GRRCF-aug-POE, F1-score: 0.603. (f) GRHED-aug-POE, F1-score: 0.7. (g) GRHED-aug-Hough-2, F1-score: 0.614. (h) GRHED-aug-Hough-3, F1-score: 0.734.

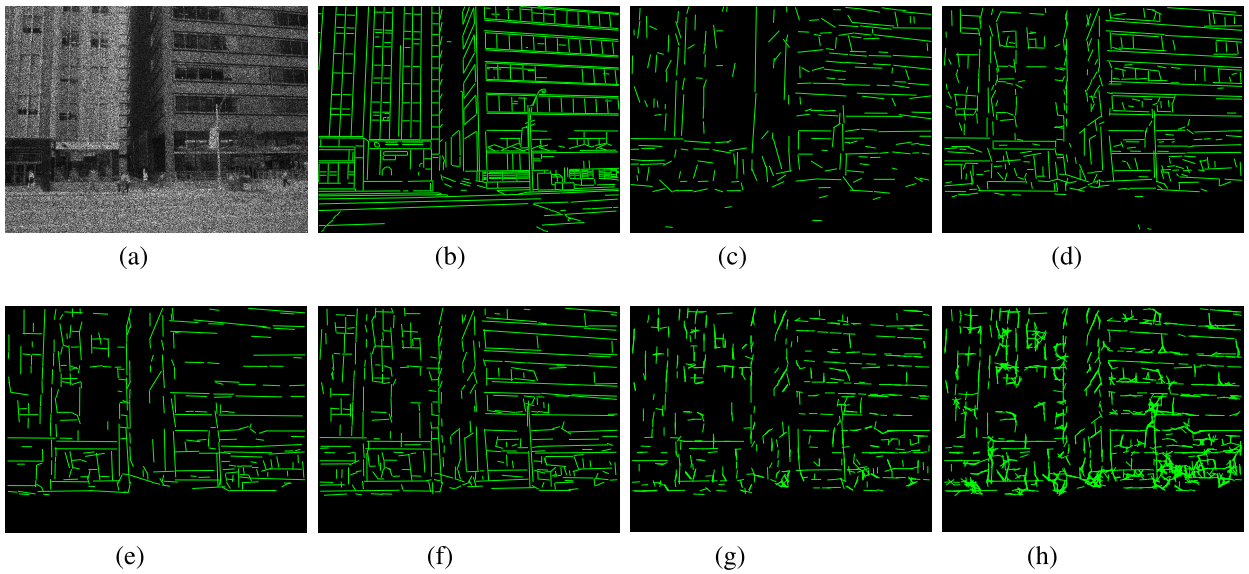


Fig. 10. Line segment detection results obtained by Hough transform-based method, LSDSAR and POE-based line segment detectors in a 1-look simulated SAR image (outdoor). Size of the image is 480 × 640 pixels. (a) 1-look image. (b) Ground truth. (c) LSDSAR ($\alpha = 4$), F1-score: 0.494. (d) GRHED-POE, F1-score: 0.692. (e) GRRCF-aug-POE, F1-score: 0.627. (f) GRHED-aug-POE, F1-score: 0.684. (g) GRHED-aug-Hough-2, F1-score: 0.561. (h) GRHED-aug-Hough-3, F1-score: 0.688.

line segment detection results as we detect more true line segments and less false detections. Although GRHED-aug-Hough-3 seems to obtain the highest F1-score, line segments detected by GRHED-aug-POE are more complete. For Hough transform-based methods, multiple responses exist for the same line segment and detected line segments tend to be fragmented. Besides, many true line segments are missing for GRHED-aug-Hough-2 while too many false line segments are detected by GRHED-aug-Hough-3. It is also shown from Fig. 9(g) and (h) that the performances of Hough transform-based methods are sensitive to the choice of thresholds. Similar conclusions can be obtained from Fig. 10(f)–(h).

5) Influence of Spatial Correlations on the Performances of Line Segment Detectors for SAR Images: Indeed, spatial correlations exist among neighboring pixels of real SAR images. For 1-look complex data, these correlations are mainly introduced by two factors: over-sampling and weighting of the azimuth and range spectrum [37], [38]. However, these factors can change depending on the data providers, even for images with similar resolutions. For instance, the weighting functions of TerraSAR-X and CSK images are different [39].

In order to show the influence of spatial correlations on the performances of different line segment detectors, we simulate three 1-look synthetic images with different amplitude ratio contrast of the edge and a 1-look SAR dataset

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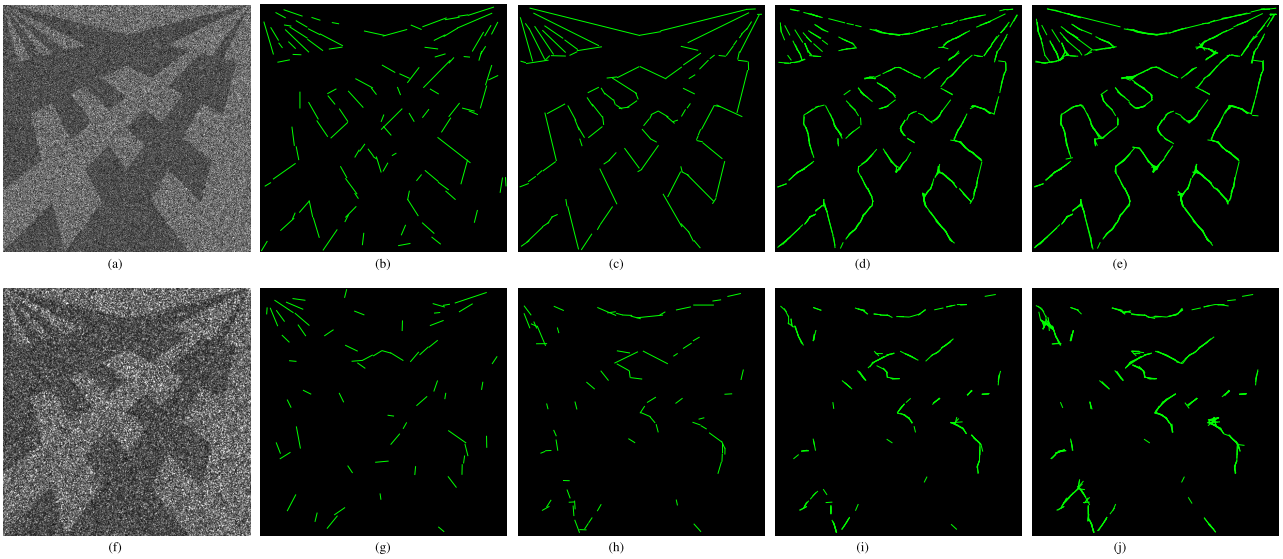


Fig. 11. Line segment detection results obtained by LSDSAR, GRHED-aug-POE, GRHED-aug-Hough-2, and GRHED-aug-Hough-3 in a 1-look synthetic SAR image with the amplitude ratio contrast of the edge 1.4. Size of the image is 512 × 512 pixels. (a) 1-look image with uncorrelated speckle noise. (b) LSDSAR, F1-score: 0.782. (c) GRHED-aug-POE, F1-score: 0.909. (d) GRHED-aug-Hough-2, F1-score: 0.782. (e) GRHED-aug-Hough-3, F1-score: 0.715. (f) 1-look image with correlated speckle noise. (g) LSDSAR, F1-score: 0.423. (h) GRHED-aug-POE, F1-score: 0.459. (i) GRHED-aug-Hough-2, F1-score: 0.462. (j) GRHED-aug-Hough-3, F1-score: 0.464.

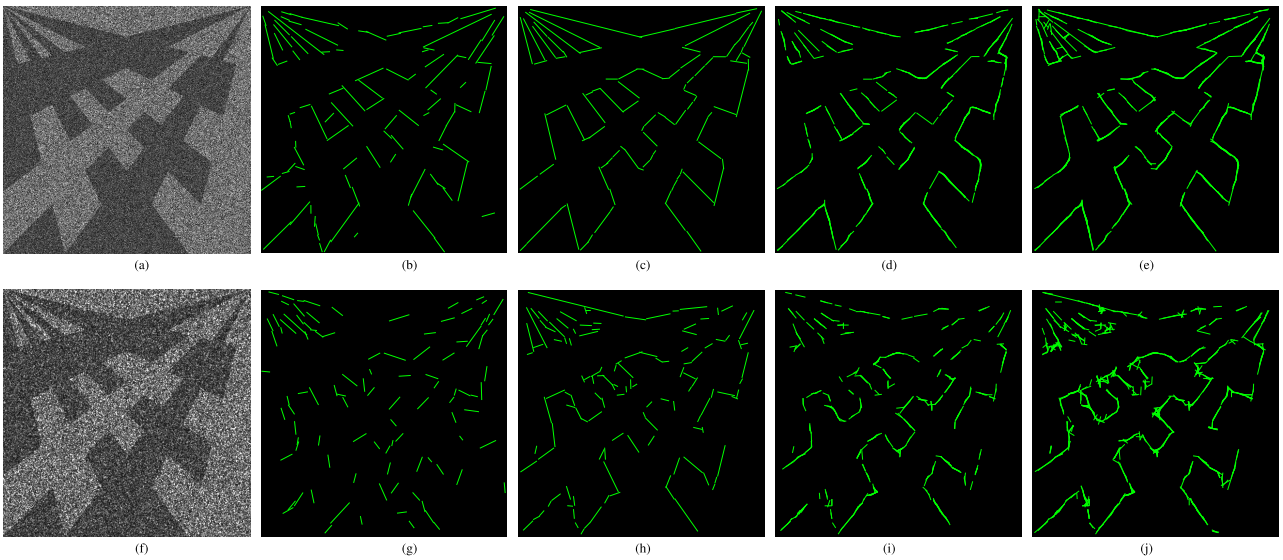


Fig. 12. Line segment detection results obtained by LSDSAR, GRHED-aug-POE, GRHED-aug-Hough-2, and GRHED-aug-Hough-3 in a 1-look synthetic SAR image with the amplitude ratio contrast of the edge 1.6. Size of the image is 512 × 512 pixels. (a) 1-look image with uncorrelated speckle noise. (b) LSDSAR, F1-score: 0.895. (c) GRHED-aug-POE, F1-score: 0.946. (d) GRHED-aug-Hough-2, F1-score: 0.811. (e) GRHED-aug-Hough-3, F1-score: 0.760. (f) 1-look image with correlated speckle noise. (g) LSDSAR, F1-score: 0.615. (h) GRHED-aug-POE, F1-score: 0.809. (i) GRHED-aug-Hough-2, F1-score: 0.720. (j) GRHED-aug-Hough-3, F1-score: 0.607.

using YorkUrban-LineSegment dataset. The spatially correlated speckle noise is simulated using the parameters of TerraSAR-X images. In StripMap mode, the spatial correlations of the speckle noise mainly come from two points: apodization applied after synthesis and image oversampling by zero-padding interpolation. Refer to [39], [40], [41] for more details about these steps.

We simulate three synthetic images of size 512 × 512 pixels with the amplitude ratio contrast of the edge 1.4, 1.6, and 1.8. These images are corrupted by both uncorrelated speckle and correlated speckle noise to produce the simulated SAR images. We compare the performances of LSDSAR with $\alpha = 4$, GRHED-aug-POE with $\rho = 0.4$, GRHED-aug-Hough-2 with $\rho = 0.4$ and GRHED-aug-Hough-3 with $\rho = 0.4$ in these simulated SAR images. Line segment detection results produced by them are displayed in Figs. 11–13. It is clearly shown from these figures that GRHED-aug-POE obtains consistently better line segment detection results than LSDSAR and Hough transform-based methods. Our POE-based method detects more true line segments with very few false detections. For Hough transform-based methods, multiple responses exist for the same line segment. GRHED-aug-Hough-2 detects fewer

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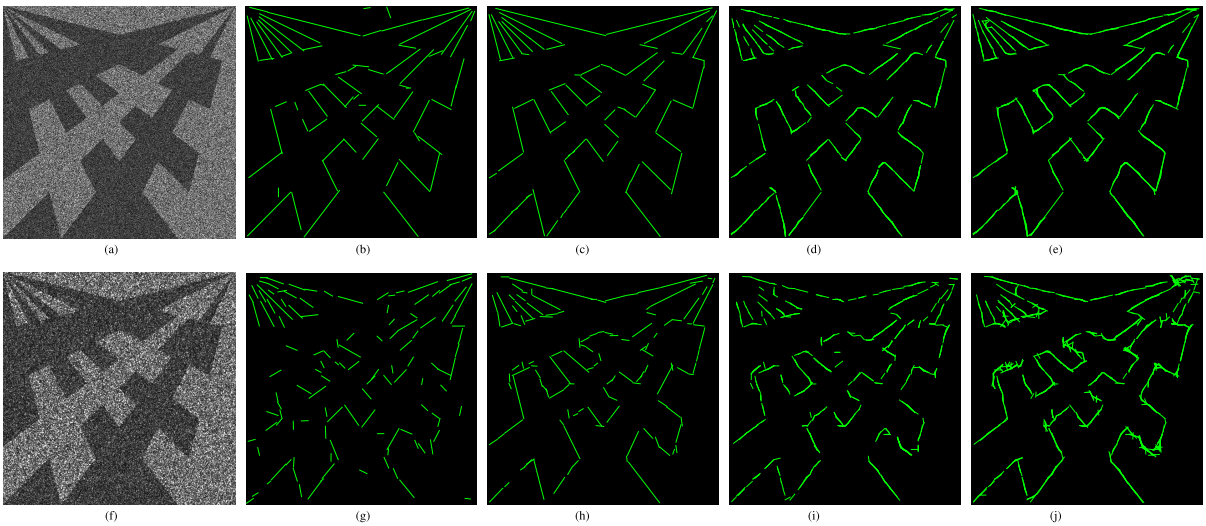


Fig. 13. Line segment detection results obtained by LSDSAR, GRHED-aug-POE, GRHED-aug-Hough-2, and GRHED-aug-Hough-3 in a 1-look synthetic SAR image with the amplitude ratio contrast of the edge 1.8. Size of the image is 512 × 512 pixels. (a) 1-look image with uncorrelated speckle noise. (b) LSDSAR, F1-score: 0.945. (c) GRHED-aug-POE, F1-score: 0.951. (d) GRHED-aug-Hough-2, F1-score: 0.822. (e) GRHED-aug-Hough-3, F1-score: 0.760. (f) 1-look image with correlated speckle noise. (g) LSDSAR, F1-score: 0.804. (h) GRHED-aug-POE, F1-score: 0.883. (i) GRHED-aug-Hough-2, F1-score: 0.784. (j) GRHED-aug-Hough-3, F1-score: 0.642.

true line segments while GRHED-aug-Hough-3 detects more false line segments. The F1-scores obtained by GRHED-aug-POE are usually much higher than those obtained by LSDSAR, GRHED-aug-Hough-2, and GRHED-aug-Hough-3. Besides, the differences in the performances of GRHED-aug-Hough-2 and GRHED-aug-Hough-3 show that the performances of Hough transform-based method are very sensitive to the choice of parameter settings.

It can also be seen from Figs. 11–13 that the spatial correlations of speckle noise decrease the performances of all the line segment detectors. All these methods are much more efficient in images with uncorrelated speckle noise. This is reasonable as all these methods assume spatial independence of the speckle noise.

We also simulate a SAR dataset using YorkUrban-LineSegment dataset by corrupting the images with spatially correlated speckle noise, and compare the performances of LSDSAR (with $\alpha = 4$), GRHED-aug-POE (with $\rho = 0.4$), GRHED-aug-Hough-2 (with $\rho = 0.4$), and GRHED-aug-Hough-3 (with $\rho = 0.4$) on it. The F1-scores obtained by these methods in the simulated dataset are displayed in Table VIII. It can be seen from the first row of Table VIII that the F1-score obtained by GRHED-aug-POE is around 12% higher than that of LSDSAR and around 6% higher than that of GRHED-aug-Hough-2, However, the F1-score computed by GRHED-aug-POE is around 7% lower than that of GRHED-aug-Hough-3. It is also shown in Table VIII that the F1-scores obtained by these methods in the images corrupted by spatially correlated speckle noise are much lower (at least 7% lower) than those obtained by them in the dataset simulated using uncorrelated speckle noise. Despite the fact that GRHED-aug-Hough-3 obtains higher F1-scores than GRHED-aug-POE, GRHED-aug-Hough-3 detects more false line segments and produces multiple responses for the same line segment, as can be seen from Fig. 14 where the line segment detection results computed by LSDSAR, GRHED-aug-POE, GRHED-aug-Hough-2, and GRHED-aug-Hough-3 in a 1-look simulated SAR image with spatially correlated noise are displayed. It is clearly shown in Fig. 14 that our proposed method GRHED-aug-POE is

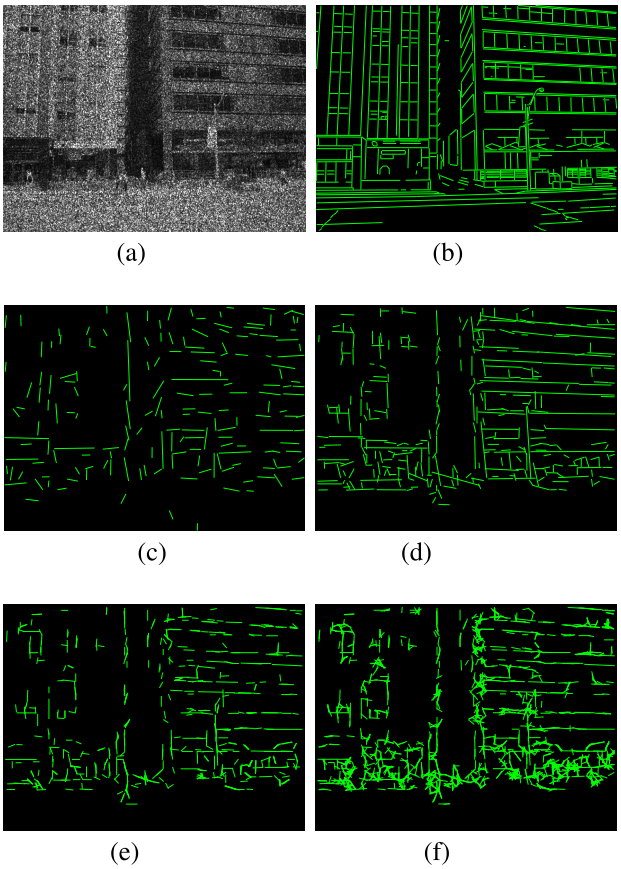


Fig. 14. Line segment detection results obtained by LSDSAR, GRHED-aug-POE, GRHED-aug-Hough-2, and GRHED-aug-Hough-3 in a 1-look simulated SAR image with spatially correlated speckle noise (outdoor). Size of the image is 480 × 640 pixels. (a) 1-look image with correlated speckle noise. (b) Ground truth. (c) LSDSAR ($\alpha = 4$), F1-score: 0.347. (d) GRHED-aug-POE, F1-score: 0.569. (e) GRHED-aug-Hough-2, F1-score: 0.493. (f) GRHED-aug-Hough-3, F1-score: 0.621.

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TABLE VIII

F1-SCORES OBTAINED BY LSDSAR WITH $\alpha = 4$, GRHED-AUG-POE WITH $\rho = 0.4$, GRHED-AUG-HOUGH-2 WITH $\rho = 0.4$ AND GRHED-AUG-HOUGH-3 WITH $\rho = 0.4$ IN THE 1-LOOK YORKURBAN-LINESEGMENT DATASETS WHICH ARE CORRUPTED BY CORRELATED AND UNCORRELATED SPECKLE NOISE

Methods	LSDSAR	GRHED-aug-POE	GRHED-aug-Hough-2	GRHED-aug-Hough-3
F1-score (correlated)	0.3809	0.5028	0.4401	0.5767
F1-score (uncorrelated)	0.4810	0.6178	0.5283	0.6445

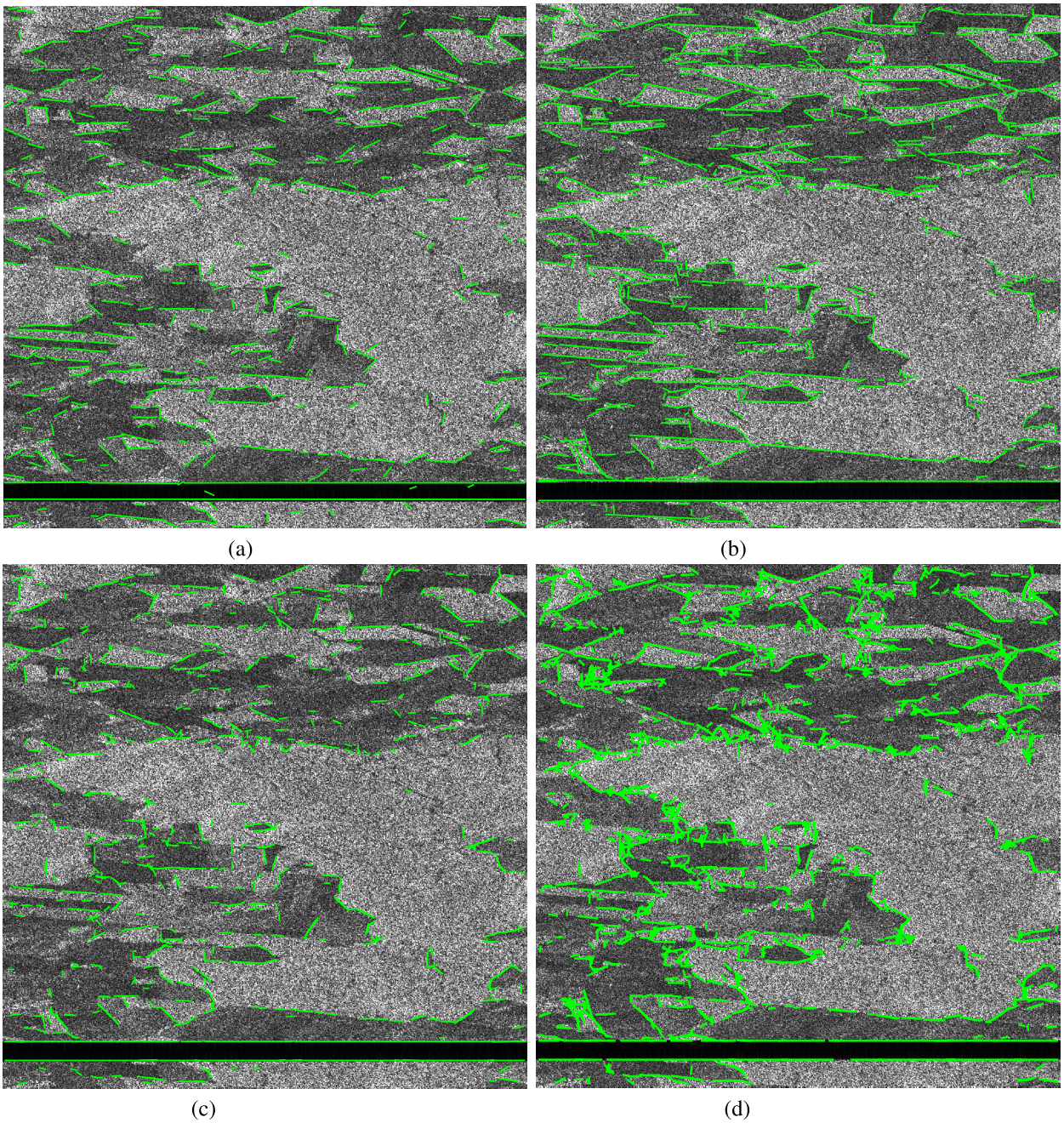


Fig. 15. Line segment detection results (shown in green lines) computed by Hough transform-based method, LSDSAR, and the POE-based method in a 1-look Sentinel-1 image (Paris, France) of size 1024 × 1024 pixels. (a) LSDSAR. (b) GRHED-aug-POE. (c) GRHED-aug-Hough-2. (d) GRHED-aug-Hough-3.

much more efficient than the other methods. More true line segments are detected with only a few false detections. The F1-score obtained by GRHED-aug-POE is around 22% higher than that of LSDSAR and 7% higher than that of GRHED-aug-Hough-2. Although the F1-score computed by GRHED-aug-Hough-3 is around 5% higher than that of GRHED-aug-POE, GRHED-aug-Hough-3 detects much more false line segments. Besides, multiple responses exist for the same line segment for GRHED-aug-Hough-3. Furthermore, the results computed by GRHED-aug-Hough-2

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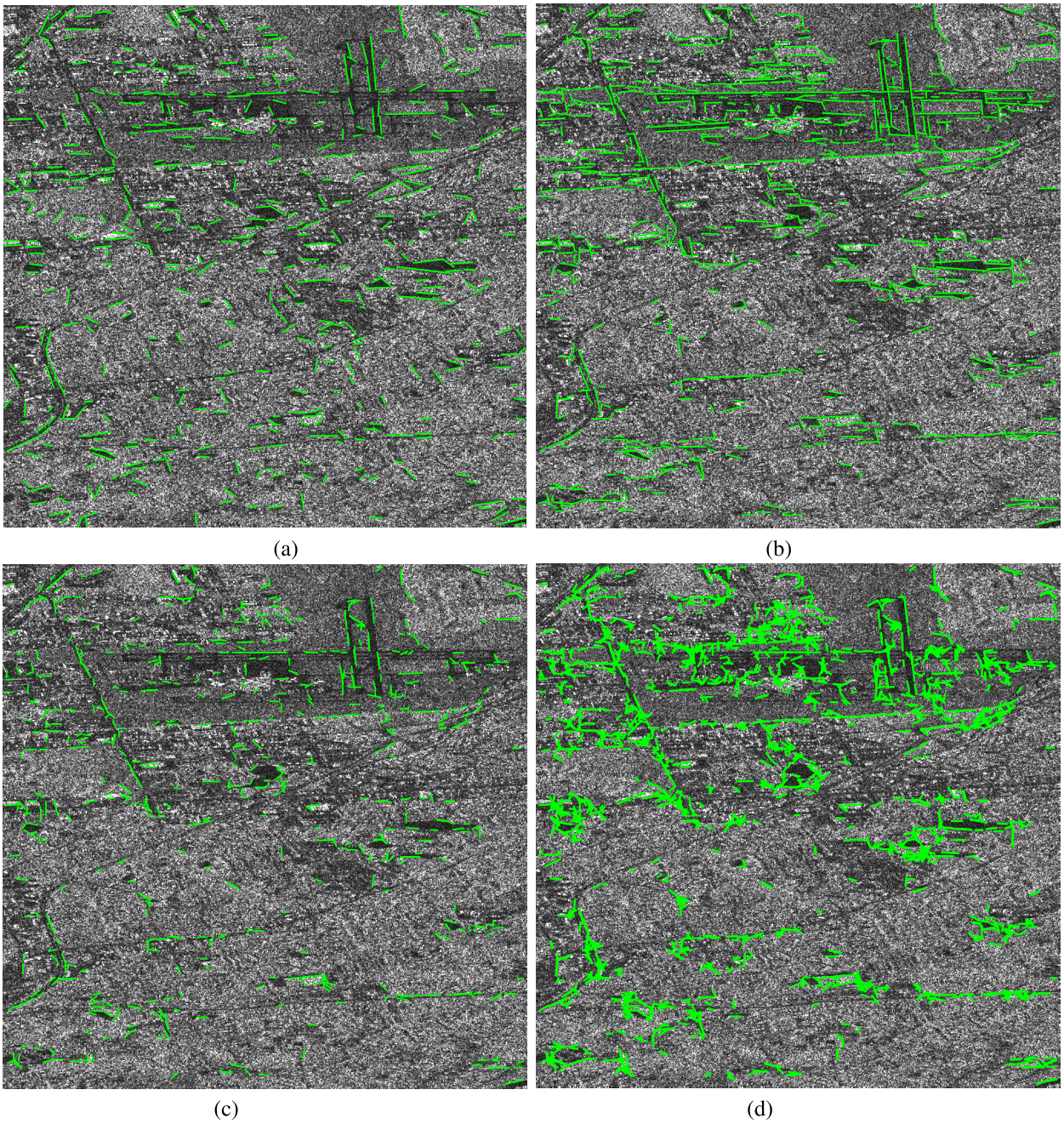


Fig. 16. Line segment detection results (shown in green lines) computed by Hough transform-based method, LSDSAR, and the POE-based method in a 1-look Sentinel-1 image (Toronto, Canada) of size 1024 × 1024 pixels. (a) LSDSAR. (b) GRHED-aug-POE. (c) GRHED-aug-Hough-2. (d) GRHED-aug-Hough-3.

and GRHED-aug-Hough-3 show two drawbacks of the Hough transform-based methods: 1) they produce multiple responses for the same line segment and 2) their performances are sensitive to the choice of parameter settings.

LSDSAR uses a ratio operator to compute the local orientations of pixels, while the justification of the ratio operators for SAR images relies on the assumption of uncorrelated speckle noise [17]. GRHED-aug-POE, GRHED-aug-Hough-2, and GRHED-aug-Hough-3 also contain a ratio operator, and the edge detector GRHED-aug is trained using images corrupted by spatially uncorrelated speckle noise. Therefore, it is reasonable to see the decrease of performances when they are tested in images contaminated by spatially correlated speckle noise. As almost all line segment detectors and edge detectors for SAR images depend on the ratio operation, all of them assume spatial independence of the speckle noise. Considering the decrease of performances of LSDSAR, GRHED-aug-POE, GRHED-aug-Hough-2, and GRHED-aug-Hough-3, we will work on how to tackle the spatial correlations of real SAR images in the future (some research studies dedicated to speckle decorrelation of real SAR images can be found in [40], [41], [42], [43], and [44]).

B. Testing in Three 1-Look Real SAR Images

The efficiency of the proposed line segment detection framework has been demonstrated in simulated SAR images, but it is still necessary to check its efficiency in real SAR images. We compare the performances of Hough transform-based

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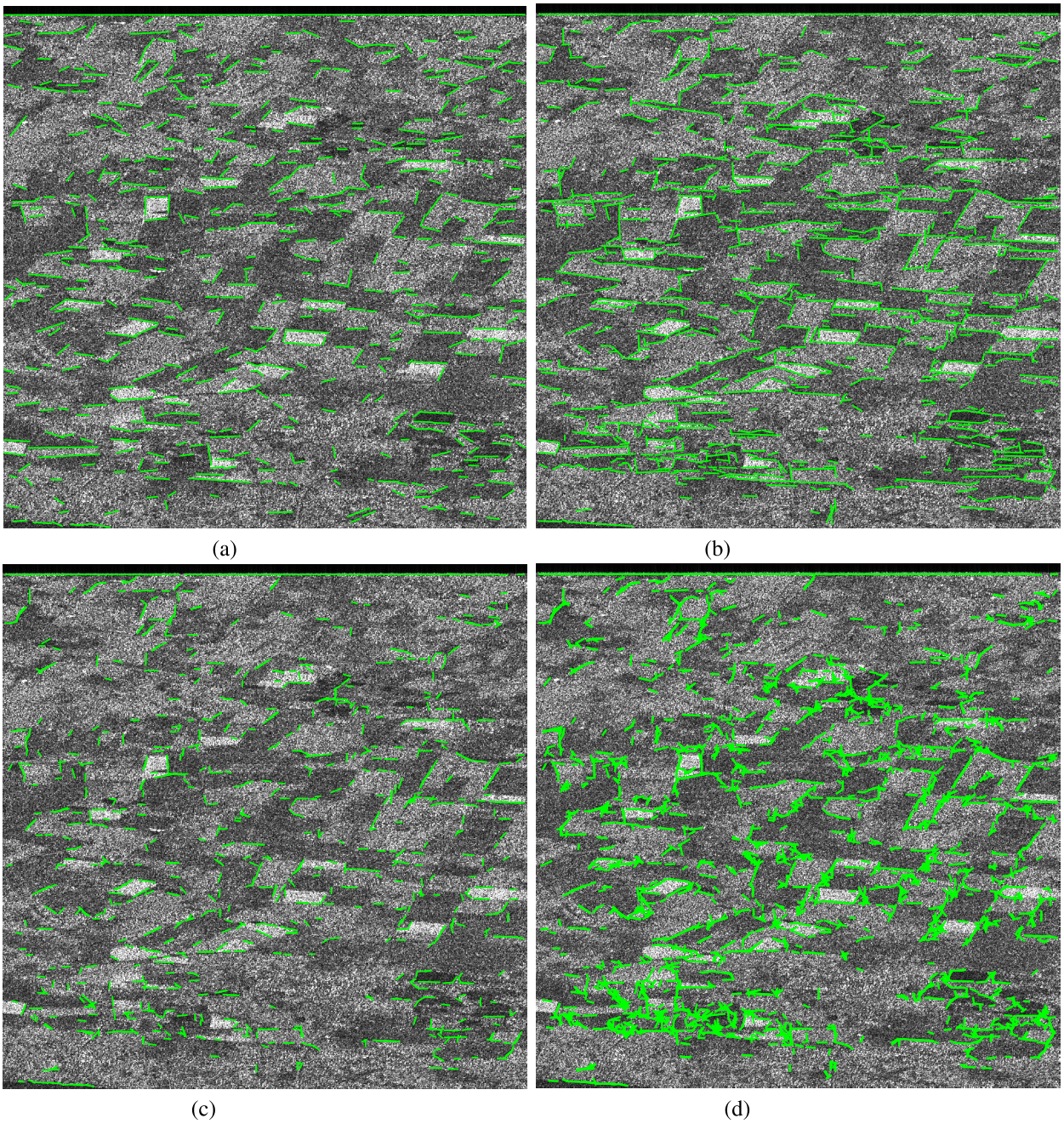


Fig. 17. Line segment detection results (shown in green lines) computed by Hough transform-based method, LSDSAR, and the POE-based method in a 1-look Sentinel-1 image (Lelystad, Netherlands) of size 1024 × 1024 pixels. (a) LSDSAR. (b) GRHED-aug-POE. (c) GRHED-aug-Hough-2. (d) GRHED-aug-Hough-3.

method, LSDSAR, and our POE-based method in three Sentinel-1 (ESA) single look complex images of different scenes. The size of the image is 1024 × 1024 pixels. The resolution ($rg \times az$) of the images are from 2.7 × 22 m to 3.5 × 22 m. Pixel spacing ($rg \times az$) are 2.3 × 14.1 m. More details about these images can be found in Table IX.

Edge maps computed by GRHED-aug with $\rho = 0.4$ are used as the input of Hough transform and our POE-based method. We use the default parameter settings in LSDSAR with $\alpha = 4$. Two sets of parameter values are used for Hough transform and we use the proposed setting for our POE-based method. Line segment detection results computed by different methods in the three 1-look Sentinel-1 images are shown in Figs. 15–17. It is clearly shown from all these figures that the proposed POE-based line segment detector detects much more true line segments than the other methods while detecting only a few false line segments. In addition, line segments detected by GRHED-aug-POE are more complete compared to those detected by LSDSAR and GRHED-aug-Hough-2.

TABLE IX

THREE SENTINEL-1 SINGLE LOOK COMPLEX IMAGES THAT ARE USED TO COMPARE THE PERFORMANCES OF DIFFERENT METHODS

Area	Number of looks	mode	date
Paris, France	1	IW mode	May 12, 2021
Toronto, Canada	1	IW mode	June 27, 2021
Lelystad, Netherlands	1	IW mode	June 29, 2021

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The proposed POE-based method fully exploits the efficiency of the deep learning-based edge detector (GRHED-aug) to emphasize true edge pixels and suppress false edge pixels. Due to the inefficiency of Hough transform to group edge pixels, it detects either a small number of fragmented line segments or too many false line segments, as can be seen from the second row of all these figures. In the meantime, the new edge grouping strategy based on our POE method shows its efficiency. GRHED-aug-POE fully exploits the detection capability of the deep learning-based edge detector and outperforms the existing methods a lot.

V. Conclusion

In this work, we propose a novel POE method, which is able to estimate the orientation of pixels in binary edge maps. On the basis of POE, we propose a new line segment detection framework that can be combined with any edge detectors. Compared to Hough transform, which is the most commonly used method to detect line segments from binary edge maps, the benefits of the proposed method are clear: first, few parameter tunings are required in POE-based methods; second, the endpoints of line segments will be naturally identified upon the generation of candidate line segments, while in Hough transform, a threshold has to be set and it is difficult to choose one that works well globally; third, as line segments are described with rectangles, we do not face the problem of multiple responses faced by Hough transform. In addition, the proposed line segment detection framework enables us to leverage the success of deep learning models for edge detection in SAR images. The efficiency of the proposed line segment detection framework is demonstrated in both simulated SAR and real SAR images. However, it should be noted that many false line segments could be detected if lots of false edge pixels are detected during edge detection, which means that the performances of the proposed line segment detection framework relies a lot on the performances of edge detectors. Thanks to the success of deep learning models for edge detection in SAR images, we advance the performances of the state-of-the-art method by a large margin in both simulated SAR and real SAR images.

Source Code

The source code of this work is available at https://github.com/ChenguangTelecom/POE_line_detect.

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